STRATIFICATION OF A QUASI-COMPACT ALGEBRAIC SPACE BY SCHEME PIECES VIA SYMMETRIC-GROUP QUOTIENTS

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Quasi-compact spaces have affine etale covers. Properties of Spaces, lemma quasi-compact-affine-cover (cross-chapter, named). Provides U -> X surjective etale affine.
- Diagonals of etale separated morphisms. Morphisms of Spaces, lemmas etale-unramified and diagonal-unramified-morphism (cross-chapter, named). All diagonals are open and closed immersions.
- Finite sets in affine opens. More on Morphisms, lemma separated-locally-quasi-finite-over-affine (cross-chapter, named). Every finite set of points of W_p lies in an affine open.
- Finite flat equivalence relations. Properties of Spaces, proposition finite-flat-equivalence-global (cross-chapter, named). The quotient V_p = W_p/S_p is a scheme.
- Etale descent of morphism properties and quotients. spaces-properties lemmas etale-local and quotient-separated, and Morphisms of Spaces lemma etale-universally-injective-open (cross-chapter, named). V_p -> X etale; V_p separated; Z_p -> T_p an open immersion.

Source reference. Original reference 07S0: focused-seeds/geometry/S9-decent-filter-quasi-compact/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original remark 03II: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 459–459. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03HG: focused-seeds/geometry/S9-decent-filter-quasi-compact/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05W1: focused-seeds/geometry/S9-decent-filter-quasi-compact/ref-tags.txt, source lines 7–7. Navigation pointer only: the

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Source reference. Original reference 05W5: focused-seeds/geometry/S9-decent-filter-quasi-compact/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06CR: focused-seeds/geometry/S9-decent-filter-quasi-compact/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03KN: focused-seeds/geometry/S9-decent-filter-quasi-compact/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03FS: focused-seeds/geometry/S9-decent-filter-quasi-compact/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03H6: focused-seeds/geometry/S9-decent-filter-quasi-compact/ref-tags.txt, source lines 12–12. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BBM: focused-seeds/geometry/S9-decent-filter-quasi-compact/ref-tags.txt, source lines 13–13. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07S6: focused-seeds/geometry/S9-decent-filter-quasi-compact/ref-tags.txt, source lines 14–14. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be a quasi-compact algebraic space over S. There exist open subspaces*

$$\ldots \subset U_4 \subset U_3 \subset U_2 \subset U_1 = X$$

with the following properties:

(1) *setting* $T_p = U_p \setminus U_{p+1}$ *(with reduced induced subspace structure) there exists a separated scheme* V_p *and a surjective étale morphism* $f_p : V_p \to U_p$ *such that* $f_p^{-1}(T_p) \to T_p$ *is an isomorphism,*
(2) *if* $x \in |X|$ *can be represented by a quasi-compact morphism* $\mathrm{Spec}(k) \to X$ *from a field, then* $x \in T_p$ *for some* p.

Proof. By Properties of Spaces, Lemma 03H6 we can choose an affine scheme U and a surjective étale morphism $U \to X$. For $p \geq 0$ set

$$W_p = U \times_X \ldots \times_X U \setminus \text{all diagonals}$$

where the fibre product has p factors. Since U is separated, the morphism $U \to X$ is separated and all fibre products $U \times_X \ldots \times_X U$ are separated schemes. Since $U \to X$ is separated the diagonal $U \to U \times_X U$ is a closed immersion. Since $U \to X$ is étale the diagonal $U \to U \times_X U$ is an open immersion, see Morphisms of Spaces, Lemmas 06CR and 05W1. Similarly, all the diagonal morphisms are open and closed immersions and W_p is an open and closed subscheme of $U \times_X \ldots \times_X U$. Moreover, the morphism

$$U \times_X \ldots \times_X U \longrightarrow U \times_{\mathrm{Spec}(\mathbf{Z})} \ldots \times_{\mathrm{Spec}(\mathbf{Z})} U$$

is locally quasi-finite and separated (Morphisms of Spaces, Lemma 03KN) and its target is an affine scheme. Hence every finite set of points of $U \times_X \ldots \times_X U$ is contained in an affine open, see More on Morphisms, Lemma 07S0. Therefore, the same is true for W_p. There is a free action of the symmetric group S_p on W_p over X (because we threw out the fix point locus from $U \times_X \ldots \times_X U$). By the above and Properties of Spaces, Proposition 07S6 the quotient $V_p = W_p/S_p$ is a scheme. Since the action of S_p on W_p was over X, there is a morphism $V_p \to X$. Since $W_p \to X$ is étale and since $W_p \to V_p$ is surjective étale, it follows that also $V_p \to X$ is étale, see Properties of Spaces, Lemma 03FS. Observe that V_p is a separated scheme by Properties of Spaces, Lemma 0BBM.

We let $U_p \subset X$ be the open subspace which is the image of $V_p \to X$. By construction a morphism $\mathrm{Spec}(k) \to X$ with k algebraically closed, factors through U_p if and only if $U \times_X \mathrm{Spec}(k)$ has $\geq p$ points; as usual observe that $U \times_X \mathrm{Spec}(k)$ is scheme theoretically a disjoint union of (possibly infinitely many) copies of $\mathrm{Spec}(k)$, see Remark 03II. It follows that the U_p give a filtration of X as stated in the lemma. Moreover, our morphism $\mathrm{Spec}(k) \to X$ factors through T_p if and only if $U \times_X \mathrm{Spec}(k)$ has exactly p points. In this case we see that $V_p \times_X \mathrm{Spec}(k)$ has exactly one point. Set $Z_p = f_p^{-1}(T_p) \subset V_p$. This is a closed subscheme of V_p. Then $Z_p \to T_p$ is an étale morphism between algebraic spaces which induces a bijection on k-valued points for any algebraically closed field k. To be sure this implies that $Z_p \to T_p$ is universally injective, whence an open immersion by Morphisms of Spaces, Lemma 05W5 hence an isomorphism and (1) has been proved.

Let $x : \mathrm{Spec}(k) \to X$ be a quasi-compact morphism where k is a field. Then the composition $\mathrm{Spec}(\overline{k}) \to \mathrm{Spec}(k) \to X$ is quasi-compact as well (Morphisms of Spaces, Lemma 03HG). In this case the scheme $U \times_X \mathrm{Spec}(\overline{k})$ is quasi-compact. In view of the fact (seen above) that it is a disjoint union of copies of $\mathrm{Spec}(\overline{k})$ we find that it has finitely many points. If the number of points is p, then we see that indeed $x \in T_p$ and the proof is finished. □